CAMPUS-LOST-AND-FOUND

PRAGATI ENGINEERING COLLEGE (A)

Project Documentation:

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1. Abstract :

Campus Lost & Found is a web-based application developed to help students and college staff report, search for, and recover lost belongings within a college campus. Students frequently lose personal items such as identity cards, wallets, water bottles, books, and electronic accessories. Finding these belongings can be difficult because information about lost and found items is usually shared through classroom groups, notice boards, or direct communication.

The Campus Lost & Found system provides a centralized platform where users can report lost or found items and view information about items reported by other users. The application allows users to enter details such as the item name, description, location, date, and item status. These details help other users identify and locate their belongings.

The project is developed using Python and Flask for backend operations, HTML and CSS for the user interface, and JavaScript for interactive functionality. The current prototype demonstrates the basic workflow of reporting lost or found items and browsing available item listings.

The main objective of the project is to improve communication between students and make the process of recovering lost belongings easier, faster, and more organized within the college campus.

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2. Introduction :

In a college environment, students carry various personal belongings such as books, identity cards, wallets, mobile phones, keys, and water bottles. These items may sometimes be misplaced in classrooms, laboratories, libraries, canteens, or other campus locations.

When an item is lost, students usually ask their classmates, contact faculty members, or share messages in college groups. Similarly, students who find an unidentified item may not know how to contact its owner.

This process can be time-consuming and may result in lost belongings remaining unclaimed.

Campus Lost & Found is designed to address this problem by providing a common online platform for reporting and searching for lost and found items.

The system allows users to report items they have lost or found and provides a centralized listing of available reports. By organizing item details and their locations, the application helps users identify possible matches and communicate more effectively.

The project is developed as a Python-based web application prototype with a simple and user-friendly interface.

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3. Problem Statement :

In many colleges, there is no centralized digital platform for managing lost and found belongings.

Students who lose personal items often face difficulties identifying whether their belongings have been found. Students who find items may also struggle to reach the original owners.

The existing manual process has the following limitations:

- Lost and found information is distributed across different communication channels.
- Students may not know where to report lost belongings.
- Finding a matching item can require repeated inquiries.
- Information about the location and date of an item may be incomplete.
- Found items may remain unclaimed for long periods.
- There is no organized listing of reported lost and found items.

Therefore, a simple web-based system is needed to collect and display lost and found reports in one place, helping students search for belongings and improve the chances of recovering them.

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4. Objectives of the Project :

The main objective of Campus Lost & Found is to develop a centralized web application that helps students and college staff report and recover lost belongings.

The specific objectives are:

1. To provide a common platform for reporting lost and found items.
2. To allow users to submit details of missing belongings.
3. To allow users to report items they have found on campus.
4. To display lost and found reports in an organized manner.
5. To provide search functionality for locating reported items.
6. To include item descriptions, locations, and dates to help identify belongings.
7. To reduce the time and effort required to recover lost items.
8. To improve communication between students who lose or find belongings.
9. To develop a simple and user-friendly interface using Python web technologies.

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5. Technologies Used :

The Campus Lost & Found project uses the following technologies.

5.1 Python

Python is the primary programming language used for backend development. It handles application logic, processes user requests, and manages lost and found item information.

5.2 Flask

Flask is a lightweight Python web framework used to develop the web application.

It manages routes, processes form submissions, and connects the frontend with backend functionality.

5.3 HTML

HTML is used to create the structure of the web pages, including the home page, item reporting form, and item listing page.

5.4 CSS

CSS is used to design the website and improve its appearance. It provides layouts, colors, typography, spacing, and responsive styling.

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5.5 JavaScript

JavaScript is used to add interactive features to the application, such as form validation, search interactions, and dynamic user interface behavior.

5.6 SQLite

SQLite can be used as a database for storing lost and found item details, user information, and item status. In the current prototype, sample item data may also be maintained in Python memory.

5.7 Visual Studio Code

Visual Studio Code is used as the development environment for writing, editing, and managing the project's source code.

5.8 Git and GitHub

Git is used for version control, and GitHub is used to store the source code and support collaboration among team members.

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6.System Requirements :

6.1 Hardware Requirements

The following hardware configuration is recommended for developing and running the application.

- Processor: Intel Core i3 or equivalent and above.
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: Minimum 1 GB of available space for project files and dependencies.
- Display: Standard laptop or desktop display.
- Internet connection: Required for installing dependencies and accessing the hosted application.

6.2 Software Requirements

- Operating System: Windows 10 or Windows 11.
- Programming Language: Python 3.
- Backend Framework: Flask.
- Frontend: HTML, CSS, and JavaScript.
- Database: SQLite, if persistent database storage is implemented.
- Code Editor: Visual Studio Code.
- Web Browser: Google Chrome, Microsoft Edge, or another modern browser.

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7. System Design :

The Campus Lost & Found application follows a basic web application architecture consisting of three main layers.

7.1 Presentation Layer

The presentation layer provides the interface through which users interact with the application.

It includes:

- Home page.
- Lost item reporting form.
- Found item reporting form.
- Item listing page.
- Search interface.

HTML, CSS, and JavaScript are used to develop the presentation layer.

7.2 Application Layer

The application layer handles the main operations of the system.

Flask is used to process incoming requests, validate form data, and manage item reports.

The application layer performs the following operations:

- Accepts lost and found item details.

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- Processes user form submissions.
- Retrieves reported items.
- Displays item information.
- Supports search and filtering operations.

7.3 Data Layer

The data layer stores and manages the information related to reported items.

In a database-enabled version, SQLite can store the item name, description, location, date, category, and status.

The application retrieves this information and displays it to users when they browse or search for items.

In the current prototype, sample item records may be stored in a Python data structure instead of a persistent database.

8. System Flowchart :

The following flowchart represents the basic workflow of Campus Lost & Found.

START

↓

Open Campus Lost & Found Website

↓

Select an Operation

Report a Lost Item or Report a Found Item?

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If Report a Lost Item:

→ Enter Item Details

→ Enter Lost Location and Date

→ Submit Report

→ Validate Information

→ Save Item Report

→ Display Report in Item Listing

If Report a Found Item:

→ Enter Found Item Details

→ Enter Found Location and Date

→ Submit Report

→ Validate Information

→ Save Item Report

→ Display Report in Item Listing

↓

Search or Browse Reported Items

↓

Identify a Possible Matching Item

↓

END

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To create the flowchart in Word, use **Insert → Shapes → Flowchart**. Use an oval for Start and End, rectangles for operations, and a diamond for the decision between reporting a lost item and a found item.

9. Module Description :

The Campus Lost & Found application is divided into the following modules.

9.1 Home Module

The Home Module provides the main interface of the application.

Its functions include:

- Displaying the project name and purpose.
- Providing navigation to the reporting and listing pages.
- Showing basic information about the lost and found process.
- Allowing users to select the required operation.

9.2 Lost Item Reporting Module

This module allows users to report belongings that they have lost.

Users can enter:

- Item name.
- Item description.

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- Lost location.
- Lost date.
- Item category, if supported.

After submitting the form, the system processes the information and displays the lost item report in the application.

9.3 Found Item Reporting Module

This module allows users to report belongings they have found on campus.

Users can provide the item name, description, location where it was found, and the date of discovery.

The reported information helps other students identify their belongings and locate possible matching items.

9.4 Item Listing Module

The Item Listing Module displays the available lost and found reports.

Its functions include:

- Displaying reported items.
- Showing item descriptions.
- Showing locations and dates.
- Organizing lost and found information.
- Helping users identify relevant reports.

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9.5 Search Module

The Search Module helps users locate specific belongings from the available reports.

Its functions include:

- Searching by item name.
- Searching using relevant keywords.
- Filtering lost and found reports.
- Displaying matching item information.

10. Implementation :

Campus Lost & Found is implemented as a Python-based web application using Flask.

10.1 Frontend Implementation

The frontend is developed using HTML, CSS, and JavaScript.

HTML provides the structure of the home page, reporting forms, and item listings.

CSS is used to create an attractive and user-friendly design with suitable layouts, colors, and responsive elements.

JavaScript provides interactive functionality, including search operations and basic form validation.

10.2 Backend Implementation

Python and Flask are used to develop the backend.

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Flask routes handle requests from the frontend.

The main routes include:

- / – Displays the home page.
- /report – Handles the item reporting page.
- /feed – Displays the available lost and found reports.
- /api/items – Provides access to item information through API requests.

When a user submits a lost or found report, the backend receives the form data, processes the information, and makes the report available in the item listing.

10.3 Data Management

The prototype manages lost and found item information using Python data structures.

Sample records may include a lost blue water bottle reported at the Central Library and a found black wallet reported at the Main Canteen.

A future database implementation using SQLite can provide persistent storage so that item reports remain available even after the application restarts.

10.4 Testing

The prototype can be tested using the following operations:

1. Open the application in a web browser.
2. Check whether the home page loads correctly.

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3. Submit a lost item report with valid information.
4. Submit a found item report with valid information.
5. Check whether the reports appear in the item listing.
6. Search for an item using its name.
7. Verify that incomplete form submissions are handled appropriately.

The testing process helps identify errors and verify the basic workflow of the application.

11. Future Scope :

The Campus Lost & Found application can be improved with additional features in future versions.

***User Authentication*:** Login and registration can be introduced to identify users and manage their reports.

Database Integration: SQLite or MySQL can be used to store item reports permanently.

Advanced Search: Search can be enhanced using item categories, locations, dates, and keywords.

Matching System: The application can automatically identify possible matches between lost and found reports based on item descriptions, locations, and other details.

Notification System: Users can receive notifications when a possible match is found.

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Admin Dashboard: Administrators can review reports, manage inappropriate entries, and update item statuses.

Mobile Application: A mobile-friendly version or dedicated mobile application can make the system more convenient for students.

Claim Verification: A secure process can be introduced to verify ownership before returning valuable items.

12. Conclusion:

Campus Lost & Found is a Python-based web application designed to simplify the process of reporting and recovering lost belongings within a college campus.

The project provides a centralized platform where students can report lost items, report found items, and browse available item listings. It helps organize information about belongings, including their descriptions, locations, and dates.

The prototype demonstrates the basic workflow using Python, Flask, HTML, CSS, and JavaScript. It provides a foundation for further development through database integration, user authentication, image uploads, and automatic item matching.

Campus Lost & Found has the potential to improve communication among students, reduce the time spent searching for belongings, and create a more organized and efficient campus lost and found process.

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13. References and Acknowledgement :

The Campus Lost & Found project was developed based on the practical need to provide a simple and organized way for students to report and recover lost belongings within a college campus.

The project design and implementation were guided by the following references and sources of inspiration:

1. *Campus Lost and Found Process:* The existing process of reporting lost belongings and searching for found items within the college environment served as the primary inspiration for developing this project.
2. *Python Programming Concepts:* Python programming concepts such as variables, lists, dictionaries, functions, and conditional statements were used to develop the application's backend logic and manage item information.
3. *Flask Web Application Development:* Flask concepts such as application routing, handling HTTP requests, processing form submissions, and displaying dynamic web pages were used in developing the web application.
4. *Web Development Technologies:* HTML, CSS, and JavaScript concepts were applied to create the user interface, design the reporting forms, display item listings, and implement interactive features.

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5. ***Lost and Found Management Systems*:** The general working principles of lost and found management systems were considered while designing the lost item reporting, found item reporting, and item searching functionalities.
6. ***Project Development and Testing:*** The project was developed through practical coding, interface design, testing, and improvements to the reporting and item listing workflow.

The Campus Lost & Found project combines these concepts into a college-focused web application intended to make the reporting and recovery of lost belongings more convenient and organized.

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14.Algorithm of Campus Lost & Found

1. Start the application.
2. Open the Campus Lost & Found website.
3. Display the home page.
4. Select an option: Report Lost Item, Report Found Item, or View Items.
5. If the user selects Report Lost Item, enter the item name, description, location, and date.

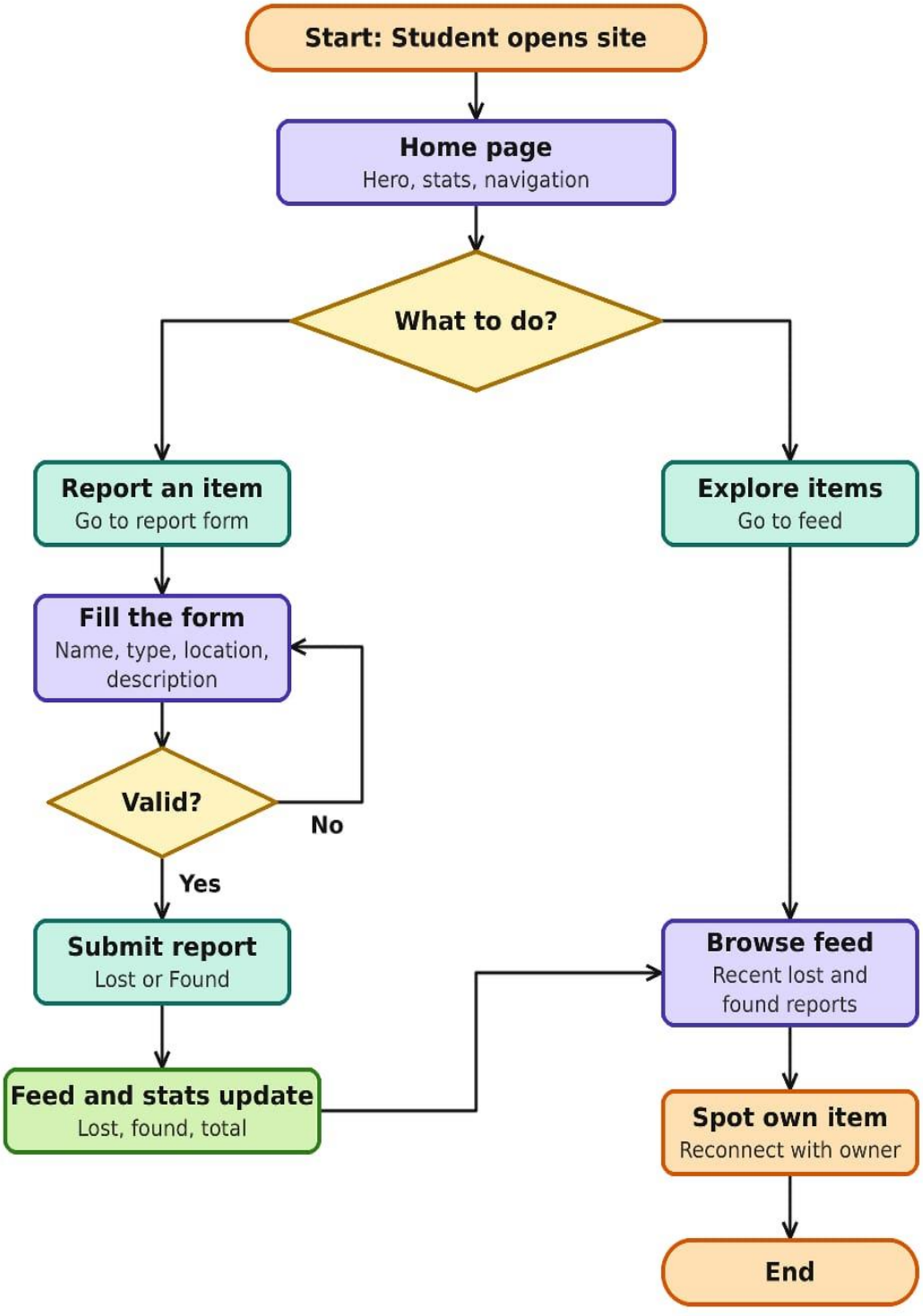
6. If the user selects Report Found Item, enter the item name, description, location, and date.
7. Check whether the required details are entered correctly.
8. If any details are missing, display a message asking the user to enter the required information.
9. If the details are complete, submit the report through the website.

Display the submitted item information on the appropriate page.

10. If the user selects View Items, display the available lost and found item information.
11. Allow the user to search for or browse the displayed items.
12. Display the relevant item information to help the user identify their belongings.
13. Stop the application.

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15.FLOWCHART:


Start: Student opens site
Home page
Hero, stats, navigation
What to do?
Report an item
Go to report form
Explore items
Go to feed
Fill the form
Name, type, location, description
Valid?
No
Yes
Submit report
Lost or Found
Feed and stats update
Lost, found, total
Browse feed
Recent lost and found reports
Spot own item
Reconnect with owner
End